

Virgo Chat Dataset Summary

Notebook

virgo-chat-dataset(1).ipynb

Phase

Chat Dataset Construction & Tokenization

Purpose

Builds the final Virgo Chat training dataset by combining multiple public conversational datasets with the custom Virgo identity dataset, followed by cleaning, quality filtering, packing and binary serialization.

Datasets Used

OpenHermes 2.5, OpenOrca, UltraChat 200K, WildChat, UltraFeedback, Magpie and the custom Virgo Identity dataset.

Dataset Limits

OpenHermes: 500k, OpenOrca: 400k, UltraChat: 400k, WildChat: 300k, UltraFeedback: 200k, Magpie: 200k samples. Virgo Identity conversations are repeated 20× to strengthen assistant identity.

Conversation Processing

Converts all datasets into a unified System/User/Assistant format before validation and tokenization.

Quality Control

Removes invalid conversations, cleans HTML and Unicode, replaces ChatGPT/Claude/Gemini references with Virgo, merges repeated roles, removes duplicates using SHA-256 hashing and scores conversation quality.

Tokenizer

Custom Virgo BPE tokenizer (45,000 vocabulary) with BOS/EOS tokens and role prefixes (<|system|>, <|user|>, <|assistant|>).

Dataset Split

Training: 99% of conversations. Validation: 1% of conversations, selected randomly during preprocessing.

Sequence Packing

Token sequences are packed into fixed 1024-token blocks and stored as uint16 binary files for efficient pretraining.

Outputs

train.bin, val.bin and checkpoint.json for resumable dataset generation.

Engineering Highlights

Streaming datasets, deterministic random seed (42), checkpointing, packed sequences, binary serialization, scalable preprocessing and identity alignment.

Summary

This notebook creates the final Virgo Chat corpus by unifying multiple instruction datasets into a clean, identity-aligned and training-ready binary dataset.